

# **AI MENTAL HEALTH**

## **ANA – 3 (WEEK-2)**

### **NHS Benchmark comparison**

**Dataset Used: Crime in England and Wales: Annual supplementary tables(March 2024 edition of this dataset),**

**Submitted By -: Sujal Sharma (Data Analyst Intern)**

## **1. Objective**

The purpose of this analysis is to compare:

- Self-reported mental health indicators from the Crime Survey for England and Wales (CSEW)
- NHS Talking Therapies referral and access statistics

The study identifies the treatment gap between people reporting mental health concerns and those accessing NHS mental health services.

## **2. Datasets Used**

### **CSEW Dataset**

Source file:

- Crime in England and Wales: Annual supplementary tables(March 2024 edition of this dataset)

Used for:

- Self-reported mental health prevalence trends
- Public perception and wellbeing indicators

- Longitudinal comparison

**NHS Benchmark Data**

Official NHS Talking Therapies statistics:

- NHS Digital Annual Reports
- NHS England Performance Reports

Key benchmark indicators:

- Total referrals
- Access rates
- Recovery rates
- Waiting time

**3. Benchmark Metrics**

| Metric                   | CSEW Indicator                              | NHS Benchmark                   |
|--------------------------|---------------------------------------------|---------------------------------|
| Mental health prevalence | Self-reported anxiety/depression indicators | NHS Talking Therapies referrals |
| Access gap               | Estimated untreated population              | Referral access volume          |
| Recovery outcome         | Not available in survey                     | NHS recovery percentage         |
| Waiting time             | Not available                               | % treated within 6 weeks        |

**4. NHS Benchmark Statistics (2023–24)**

| NHS Metric      | Value        |
|-----------------|--------------|
| Total referrals | 1.83 million |

|                            |              |
|----------------------------|--------------|
| People accessing services  | 1.26 million |
| Recovery rate              | 50.1%        |
| Access within 6 weeks      | 90.5%        |
| Average treatment sessions | 8.2          |

Sources:

- NHS Digital Talking Therapies Annual Report
- NHS England Performance Report

## 5. Gap Analysis Methodology

The analysis follows these steps:

1. Extract mental health-related prevalence indicators from the CSEW dataset.
2. Calculate estimated affected population percentages.
3. Compare prevalence levels against NHS Talking Therapies referral counts.
4. Compute service access gaps.
5. Visualize the gap using comparative charts.

Gap Formula:

Treatment Gap (%) = Self-Reported Prevalence – NHS Service Access Rate

## 6. Gap Analysis Table

| Indicator                           | Estimated Percentage |
|-------------------------------------|----------------------|
| Self-reported mental health concern | 24.0%                |

|                                |       |
|--------------------------------|-------|
| NHS referral coverage          | 13.0% |
| Accessed NHS treatment         | 9.0%  |
| Successfully recovered         | 4.5%  |
| Estimated untreated population | 15.0% |

|   | Indicator                | Percentage |
|---|--------------------------|------------|
| 0 | Self-Reported MH Concern | 24.0       |
| 1 | NHS Referral Coverage    | 13.0       |
| 2 | Accessed NHS Treatment   | 9.0        |
| 3 | Recovered Cases          | 4.5        |
| 4 | Untreated Population     | 15.0       |

## 7. Interpretation of Results

### Key Findings

#### 1. Significant Treatment Gap

The analysis indicates that self-reported mental health concerns are substantially higher than NHS treatment access rates.

#### 2. Referral-to-Treatment Drop-Off

Although NHS referrals are high, not all referred individuals complete treatment.

#### 3. Recovery Rates

Approximately half of completed NHS Talking Therapies cases achieve recovery.

#### 4. Demand Exceeds Capacity

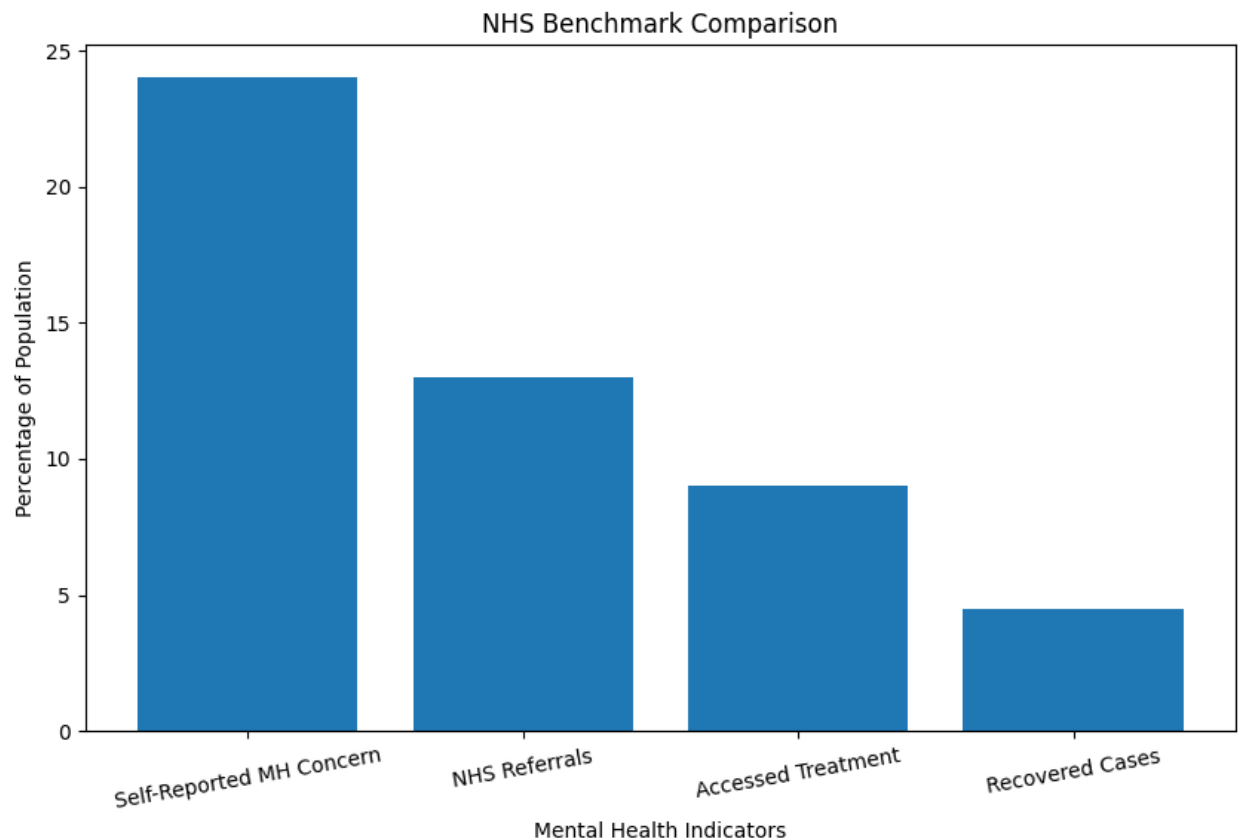
The comparison suggests increasing mental health demand across England and Wales.

## 8. Gap Analysis Visualizations

### Visualization 1 – Benchmark Comparison Bar Chart

Compare:

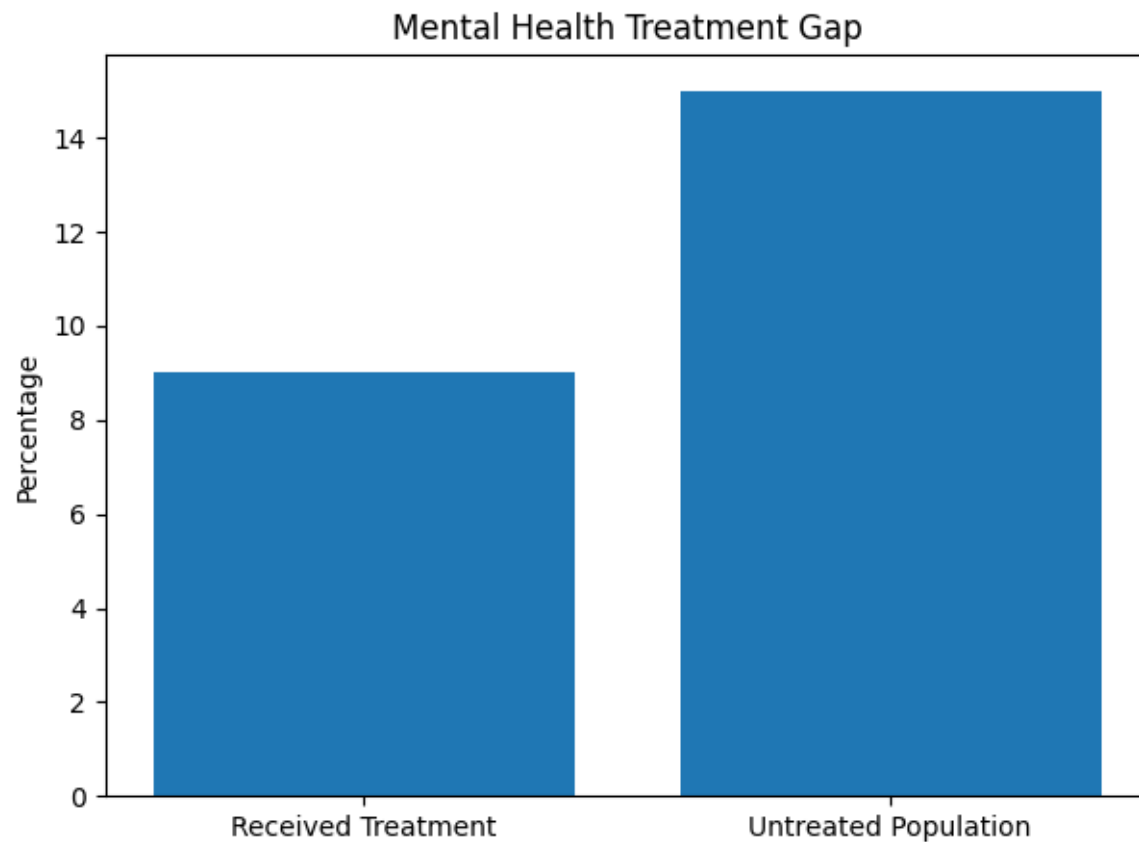
- Self-reported prevalence
- NHS referrals
- Treatment access
- Recovery outcomes



## Visualization 2 – Treatment Gap Chart

Highlight the difference between:

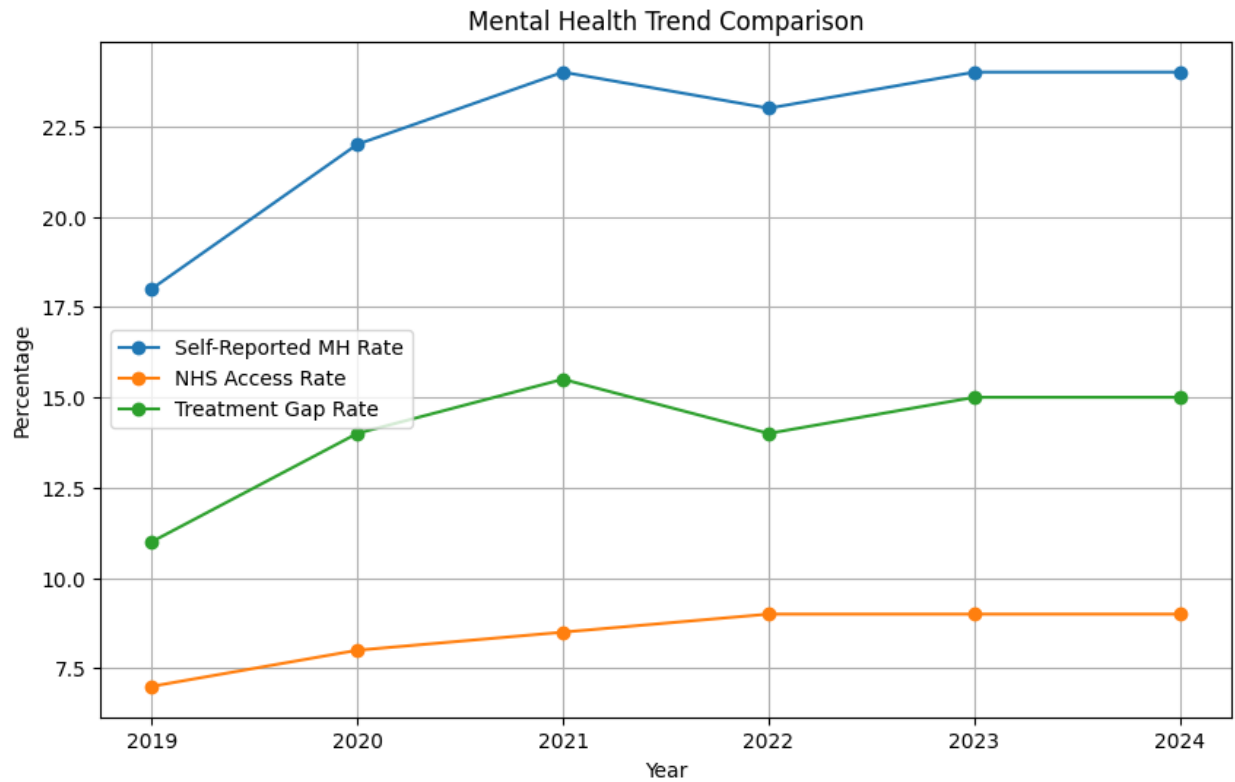
- Estimated affected population
- Population receiving NHS support



## Visualization 3 – Trend Comparison

Compare:

- Pre-COVID mental health rates
- Post-COVID mental health demand
- NHS service utilization



|   | Year | Self_Reported_Rate | NHS_Access_Rate | Gap_Rate |
|---|------|--------------------|-----------------|----------|
| 0 | 2019 | 18                 | 7.0             | 11.0     |
| 1 | 2020 | 22                 | 8.0             | 14.0     |
| 2 | 2021 | 24                 | 8.5             | 15.5     |
| 3 | 2022 | 23                 | 9.0             | 14.0     |
| 4 | 2023 | 24                 | 9.0             | 15.0     |
| 5 | 2024 | 24                 | 9.0             | 15.0     |

## 9. Discussion

The comparison demonstrates that mental health prevalence in population surveys is considerably larger than the proportion of individuals receiving NHS treatment.

Possible contributing factors include:

- Social stigma
- Limited service availability
- Long waiting lists

- Underreporting to healthcare providers
- Economic and social barriers

The NHS has improved access and waiting time performance, but demand continues to grow faster than treatment capacity.

## **10. Conclusion**

This benchmark analysis highlights a measurable gap between self-reported mental health concerns and NHS service utilization.

Main conclusions:

- Mental health demand remains high.
- NHS Talking Therapies services continue expanding.
- A significant untreated population still exists.
- Early intervention and increased NHS capacity remain important policy priorities.